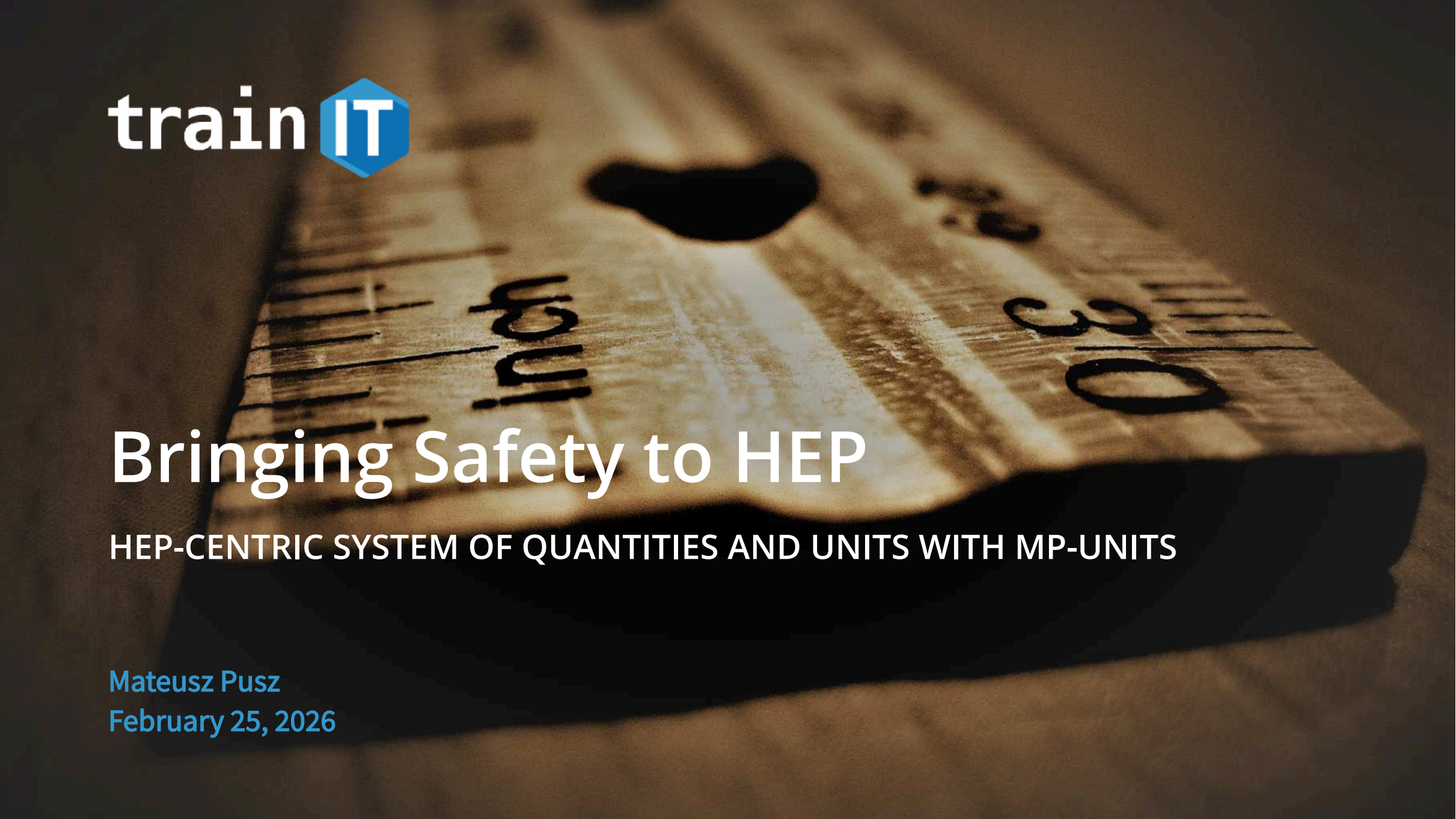

Bringing Safety to HEP

HEP-CENTRIC SYSTEM OF QUANTITIES AND UNITS WITH MP-UNITS

Mateusz Pusz

February 25, 2026

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- Hacking C++ for more than *20 years* for fun and living

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As a freelancer, hours spent on Open Source and ISO C++ Committee work compete directly with billable time. If you find this work valuable, please consider sponsoring or engaging me for training.

Disclaimer

My expertise: C++ type systems and zero-cost abstractions.
Your expertise: HEP physics.
Together: catching bugs before they publish papers! 🎯

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LLMs helped me understand HEP domain and generate proper examples. If details need refinement, please share—that's exactly the domain knowledge we want to encode into types!

C++ Safety & Scientific Correctness

In High-Energy Physics, **Safety = Correctness and Reproducibility**.

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- We trust software with
  - Complex *detector geometries* (millions of volumes)
  - Critical *tracking and reconstruction algorithms*
  - Multi-day *simulations* on the Grid

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The Problem: A single unit mismatch in a **`double`** could silently invalidate a result or cause a 10x scale error in detector simulation.

The Challenge at CERN

- **2.5M lines of code** in ATLAS Athena framework alone
- Multiple frameworks with different unit conventions
  - **Geant4**: mm, ns, MeV, rad
  - **ROOT**: cm, s, GeV, degree
  - **CLHEP**: Different CODATA versions (2014, 2018, 2022)

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Training alone won't solve this

- Code often written by physicists, not software engineers
- Silent bugs from **`double`** proliferation
- Compiler can't help when everything is dimensionless

The "Argument Soup" Problem

GEANT4: `G4Trap` CONSTRUCTOR

```
G4Trap(const G4String& name,
       G4double pDz, G4double pTheta, G4double pPhi,
       G4double pDy1, G4double pDx1, G4double pDx2, G4double pAlp1,
       G4double pDy2, G4double pDx3, G4double pDx4, G4double pAlp2);
```

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- **12 double parameters** - which are lengths? which are angles?
- Accidentally swapping **`pDx2`** and **`pAlp1`** compiles fine

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```

- **12 double parameters** - which are lengths? which are angles?
- Accidentally swapping **`pDx2`** and **`pAlp1`** compiles fine

Result: Invalid detector geometry, caught only at runtime (maybe)

The "Argument Soup" Problem

ROOT: `TGeoMedium` CONSTRUCTOR

```
TGeoMedium(const char *name, Int_t numed, TGeoMaterial *mat,
           Double_t fieldm, Double_t tmaxfd, Double_t stemax,
           Double_t deemax, Double_t epsil, Double_t stmin);
```

The "Argument Soup" Problem

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```

- **6 `Double_t` parameters after the material**
- **`fieldm`**: magnetic field (G), **`tmaxfd`**: angle (deg), **`stemax`**: length (cm)
- **`deemax`**: energy ratio (dimensionless), **`epsil`**: (dimensionless), **`stmin`**: length (cm)

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All silently convertible to each other!

The "Argument Soup" Problem

GEANT4 PARTICLE DEFINITION

```
// G4ParticleDefinition constructor: 21 parameters!
G4ParticleDefinition(const G4String& aName,
                     G4double mass, G4double width, G4double charge,
                     G4int iSpin, G4int iParity, G4int iConjugation,
                     G4int iIsospin, G4int iIsospinZ, G4int gParity,
                     const G4String& pType, G4int lepton, G4int baryon,
                     G4int encoding, G4bool stable, G4double lifetime,
                     G4DecayTable* decaytable, G4bool shortlived,
                     const G4String& subType, G4int anti_encoding,
                     G4double magneticMoment);
```

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```

Multiple dangerous pairs of same-typed parameters:

- **`mass`**, **`width`**, **`lifetime`**, **`magneticMoment`** - all **`G4double`** (energy/time)
- **`iSpin`**, **`iParity`**, **`iConjugation`**, **`iIsospin`**, **`iIsospinZ`**, **`gParity`** - all **`G4int`**

Potential HEP Bug: Parameter Swapping

```
using namespace CLHEP;

// MeV, GeV are just 'constexpr double' scale factors!
double mass = 3.872 * GeV;  // rest energy
double width = 1.2 * MeV;   // decay width

G4ParticleDefinition("X3872", width, mass, 0, ...);
```

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```

- *Compiles and runs without error* (both are **G4double**)
- Particle rest energy = 1.2 MeV (should be 3872 MeV) - factor of **3000× wrong**!
- Decay width = 3872 MeV (should be 1.2 MeV) - **absurdly large**
- Wrong decay kinematics, invalid cross-sections
- **Caught only in physics validation (or never!)**

The "Magic 1.0" Problem

Different frameworks use different internal base units

QUANTITY	GEANT4	ROOT
Length	`1.0` = 1 mm	`1.0` = 1 cm
Time	`1.0` = 1 ns	`1.0` = 1 s
Angle	`1.0` = 1 rad	`1.0` = 1 degree
Energy	`1.0` = 1 MeV	`1.0` = 1 GeV

The "Magic 1.0" Problem

```
// Legacy Bridge Code
double radius = rootVolume->GetRmax(); // Returns 10.0 (meaning 10 cm)

// Geant4 expects mm internally
G4Tubs g4Tube("Tube", 0, radius, ...);
// Interprets as 10 mm!
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Examples from real ATLAS/CMS code

- **TGeoCone** (ROOT) vs **G4Cons** (Geant4): parameter order AND units differ
- Bridging code requires manual scaling factors
- One missing **/CLHEP::cm** → 10× error

The CODATA Drift Problem

Different CERN frameworks use different physics constants:

FRAMEWORK	CODATA VERSION	ELECTRON MASS
CLHEP (old)	2014	0.5109989461 MeV
Gaudi (old)	2014	0.5109989461 MeV
Geant4 (current)	2018	0.51099895000 MeV
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- Small differences, but **accumulate** in precision calculations
- Mixing frameworks → inconsistent physics
- **Current solution**: hope everyone uses same version
- **mp-units solution**: Compile-time namespace selection

mp-units: Design Philosophy

COMPILE-TIME SAFETY

- Correct handling of physical quantities, units, and numerical values
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- Optimized for readable compilation errors

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- Dedicated system of quantities
- CODATA version selection (2014, 2018, 2022)
- Scalar, vector, and tensor quantities
- Easy to extend for custom experiment units

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STANDARDIZATION

- Foundation for future **`std::units`**
- ISO C++ proposals P1935, P2980, P3045

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6. **Affine Space Safety** - Distinguishes points and deltas

Each level adds another layer of protection, encoding domain knowledge directly into the type system.

Level 1: Dimension Safety

The real danger: wrong arguments in function calls

```
using namespace mp_units::hep::unit_symbols;

// Function expecting distance first, then time
quantity<mm / ns> average_speed(quantity<mm> distance, quantity<ns> time);
```

```
quantity bunch_time   = 25 * ns;   // LHC bunch spacing
quantity track_length = 120 * mm;  // Silicon tracker path

quantity speed = average_speed(track_length, bunch_time);  // ✅ Correct

// auto bad1 = average_speed(bunch_time, track_length);    // ❌ Compile error: args swapped!
// auto bad2 = average_speed(track_length, 45 * MeV);      // ❌ Compile error: energy ≠ time!
```

Level 2: Unit Safety

Automatic, zero-cost unit conversions

```
using namespace mp_units::hep::unit_symbols;

quantity g4_radius = 150. * mm;            // Geant4 uses mm
quantity root_radius = g4_radius.in(cm);   // ROOT uses cm

// Automatic conversion, zero runtime cost
assert(g4_radius == root_radius);          // ✅ Compiles and passes!

// Extract value in specific unit for legacy APIs
double g4_val = g4_radius.numerical_value_in(mm);     // 150.0
double root_val = root_radius.numerical_value_in(cm); // 15.0
```

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double root_val = root_radius.numerical_value_in(cm); // 15.0
```

Eliminates the ROOT/Geant4 bridge bugs:

- No manual `* 10.0` scaling factors
- Compiler handles conversions
- If units match, conversion optimizes away completely

Level 3: Representation Safety

Preventing truncation and precision loss

```
using namespace mp_units::hep::unit_symbols;

// Raw ADC readout from calorimeter
quantity<one, std::uint16_t> raw_adc = 3500;                  // 12-bit ADC

// Calibration: ADC → voltage → energy
constexpr quantity adc_to_voltage = 2500.0 * mV / 4096;  // ADC → voltage
constexpr quantity energy_scale = 45.0 * MeV / V;        // Voltage → energy
quantity voltage = raw_adc * adc_to_voltage;             // Auto-upgrades to double
quantity energy = voltage * energy_scale;                // ~96 MeV

// ❌ ERROR: Narrowing conversion
// quantity<mV, std::uint16_t> v_int = voltage;          // Compile-time error

// ✅ CORRECT: Explicit truncation when needed
quantity<mV, std::uint16_t> v_int = voltage.force_in<std::uint16_t>(mV); // Fractional mV lost!
```

Level 4: Quantity Kind Safety

*Incompatible quantities with the **same dimension** (T) — but different (sub)kind*

```
using namespace mp_units::hep::unit_symbols;

quantity<one> calc_gamma(quantity<hep::proper_time[ns]>     tau,
                         quantity<hep::coordinate_time[ns]> t_lab);
```

```
quantity tau   = hep::proper_time(0.385 * ns);     // B⁰ meson rest-frame lifetime
quantity t_lab = hep::coordinate_time(4.2 * ns);   // observed lab time of flight

quantity gamma = calc_gamma(tau, t_lab);           // ✅ Correct
// calc_gamma(t_lab, tau);                         // ❌ args swapped — kind mismatch!
// auto sum = tau + t_lab;                         // ❌ can't add different kinds!
// bool cmp = tau < t_lab;                         // ❌ can't compare different kinds!
```

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// bool cmp = tau < t_lab;                         // ❌ can't compare different kinds!
```

- Hz vs Bq, Sv vs Gy, rad vs sr (if dimensionless)

Level 5: Quantity Safety

*Turning the 11-**`double`** soup into a self-documenting API*

```
// Each geometric axis and angle role gets its own sub-kind of hep::length / hep::angle
inline constexpr struct trap_half_z      final : quantity_spec<hep::length> {} trap_half_z;
inline constexpr struct trap_half_y      final : quantity_spec<hep::length> {} trap_half_y;
inline constexpr struct trap_half_x      final : quantity_spec<hep::length> {} trap_half_x;
inline constexpr struct trap_tilt_angle  final : quantity_spec<hep::angle>  {} trap_tilt_angle;
inline constexpr struct trap_shear_angle final : quantity_spec<hep::angle>  {} trap_shear_angle;
```

```
void MakeG4Trap(quantity<trap_half_z[mm]>       pDz,
                quantity<trap_tilt_angle[rad]>  pTheta,
                quantity<trap_tilt_angle[rad]>  pPhi,
                quantity<trap_half_y[mm]>       pDy1,
                quantity<trap_half_x[mm]>       pDx1,
                quantity<trap_half_x[mm]>       pDx2,
                quantity<trap_shear_angle[rad]> pAlp1,
                quantity<trap_half_y[mm]>       pDy2,
                quantity<trap_half_x[mm]>       pDx3,
                quantity<trap_half_x[mm]>       pDx4,
                quantity<trap_shear_angle[rad]> pAlp2);
```

```
quantity pDz    = trap_half_z(150. * mm);
quantity pDy1   = trap_half_y(25. * mm);
quantity theta = trap_tilt_angle(0. * rad);

// ❌ Level 1: mm where rad expected (pDy1)!
// MakeG4Trap(pDz, pDy1, ...);
// ❌ Level 5: pDy passed where pDz expected (both mm)!
// MakeG4Trap(pDy1, theta, ...);

// ✅ Lengths of different axes promote to
// common hep::length — safe for clearance calcs
quantity half_diag = pDz + pDy1;
static_assert(half_diag.quantity_spec == hep::length);
```

Level 6: Affine Space Safety

Distinguishing points and deltas

```
using namespace mp_units::hep::unit_symbols;

// Points: absolute positions in detector
quantity_point z_vertex = z_vertex_zero + 15.0 * mm;
quantity_point z_decay = z_vertex_zero + 20.2 * mm;

// Point - Point = Delta (displacement) ✅
quantity decay_length = z_decay - z_vertex;  // 5.2 mm

// Point + Delta = Point (new position) ✅
quantity_point z_final = z_vertex + decay_length;

// ❌ ERROR: Cannot add absolute positions
// auto bad = z_vertex + z_decay;  // Compile error!
```

Bonus: Generic Interfaces

Unit-agnostic algorithms that preserve input units

```
using namespace mp_units::hep::unit_symbols;

// Algorithm is unit-agnostic: accepts mm, cm, m, etc. for position coordinates
QuantityPointOf<hep::position_vector> auto decay_vertex_position(QuantityPointOf<hep::position_vector> auto production_z,
                                                                 QuantityOf<hep::decay_length> auto decay_length,
                                                                 QuantityOf<hep::angle> auto theta)
{
  return production_z + decay_length * hep::cos(theta);  // Returns position in same unit
}

// Call with mm (Geant4) → returns position in mm
quantity_point z_decay_g4 = decay_vertex_position(point<mm>(150), 25 * mm, 18 * hep::degree);

// Call with cm (ROOT) → returns position in cm
quantity_point z_decay_root = decay_vertex_position(point<cm>(15), 2.5 * cm, 0.314 * hep::radian);
```

Benefits: Zero conversion overhead, natural units preserved, type-safe

Bonus: Strongly-Typed Dimensionless Quantities

Preventing confusion between incompatible dimensionless values

```
inline constexpr struct efficiency final : quantity_spec<dimensionless, is_kind> {} efficiency;
inline constexpr struct branching_ratio final : quantity_spec<dimensionless, is_kind> {} branching_ratio;
inline constexpr struct event_count final : quantity_spec<dimensionless, is_kind> {} event_count;
inline constexpr struct yield final : quantity_spec<dimensionless, is_kind> {} yield;

quantity<yield[one]> compute_yield(quantity<efficiency[one]> eff,
                                   quantity<event_count[one], int> n_events);

quantity trigger_eff = efficiency(0.95);
quantity selection_eff = efficiency(0.87);
quantity BR_Hgg = branching_ratio(2.27e-3);

// ❌ ERROR: Cannot add or compare incompatible dimensionless kinds
// auto bad = trigger_eff + BR_Hgg;     // Compile error!
// bool bad2 = trigger_eff > BR_Hgg;    // Compile error!

// ✅ CORRECT: Function enforces kind at call site
quantity res = compute_yield(trigger_eff, event_count(1'000'000));  // ✅ OK
// quantity res = compute_yield(BR_Hgg, event_count(1'000'000));    // ❌ Compile error!
```

Also useful for: purity, asymmetries, χ^2/ndf, ratios, confidence levels

Bonus: Faster-than-Lightspeed Constants

`c` as a compile-time unit — never multiplied at runtime

```
using namespace mp_units::hep::unit_symbols;

// c is a named_constant unit in mp-units, not a floating-point value
quantity p = 4. * GeV / c;      // type: quantity<GeV/c>   — c stays symbolic
quantity mass = 3. * GeV / c2;  // type: quantity<GeV/c²>  — c² stays symbolic
```

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```

```
quantity E = hypot(p * c, mass * c2);   // E² = (pc)² + (mc²)²  ⟶  4² + 3² = 25  ⟶  E = 5 GeV
```

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```

```
std::println("p = {} = {::N[.5g]}", p, p.in(kg * m / s));  // "p = 4 GeV/c = 2.1377e-18 m kg/s"
std::println("m = {} = {::N[.5g]}", mass, mass.in(kg));    // "m = 3 GeV/c² = 5.348e-27 kg"
std::println("E = {} = {::N[.5g]}", E, E.in(hep::joule));  // "E = 5 GeV = 8.0109e-10 J"
```

Bonus: Faster-than-Lightspeed Constants

`c` as a compile-time unit — never multiplied at runtime

```
using namespace mp_units::hep::unit_symbols;

// c is a named_constant unit in mp-units, not a floating-point value
quantity p = 4. * GeV / c;      // type: quantity<GeV/c>   — c stays symbolic
quantity mass = 3. * GeV / c2;  // type: quantity<GeV/c²>  — c² stays symbolic
```

```
quantity E = hypot(p * c, mass * c2);   // E² = (pc)² + (mc²)²  ⟶  4² + 3² = 25  ⟶  E = 5 GeV
```

```
std::println("p = {} = {::N[.5g]}", p, p.in(kg * m / s));  // "p = 4 GeV/c = 2.1377e-18 m kg/s"
std::println("m = {} = {::N[.5g]}", mass, mass.in(kg));    // "m = 3 GeV/c² = 5.348e-27 kg"
std::println("E = {} = {::N[.5g]}", E, E.in(hep::joule));  // "E = 5 GeV = 8.0109e-10 J"
```

The **c** factors (and all other constants like **π**) cancel in the type system — identical to working in natural units, with full dimensional safety.

The HEP System: Energy as Base Quantity

Why HEP physics differs from ISQ?

The HEP System: Energy as Base Quantity

Why HEP physics differs from ISQ?

ISQ (INTERNATIONAL SYSTEM OF QUANTITIES)

- *7 base dimensions:* length, mass, time, current, temperature, amount, luminosity
- *Derived:* energy = mass × length^2 / time^2 $(M \cdot L^2 \cdot T^{-2})$

The HEP System: Energy as Base Quantity

Why HEP physics differs from ISQ?

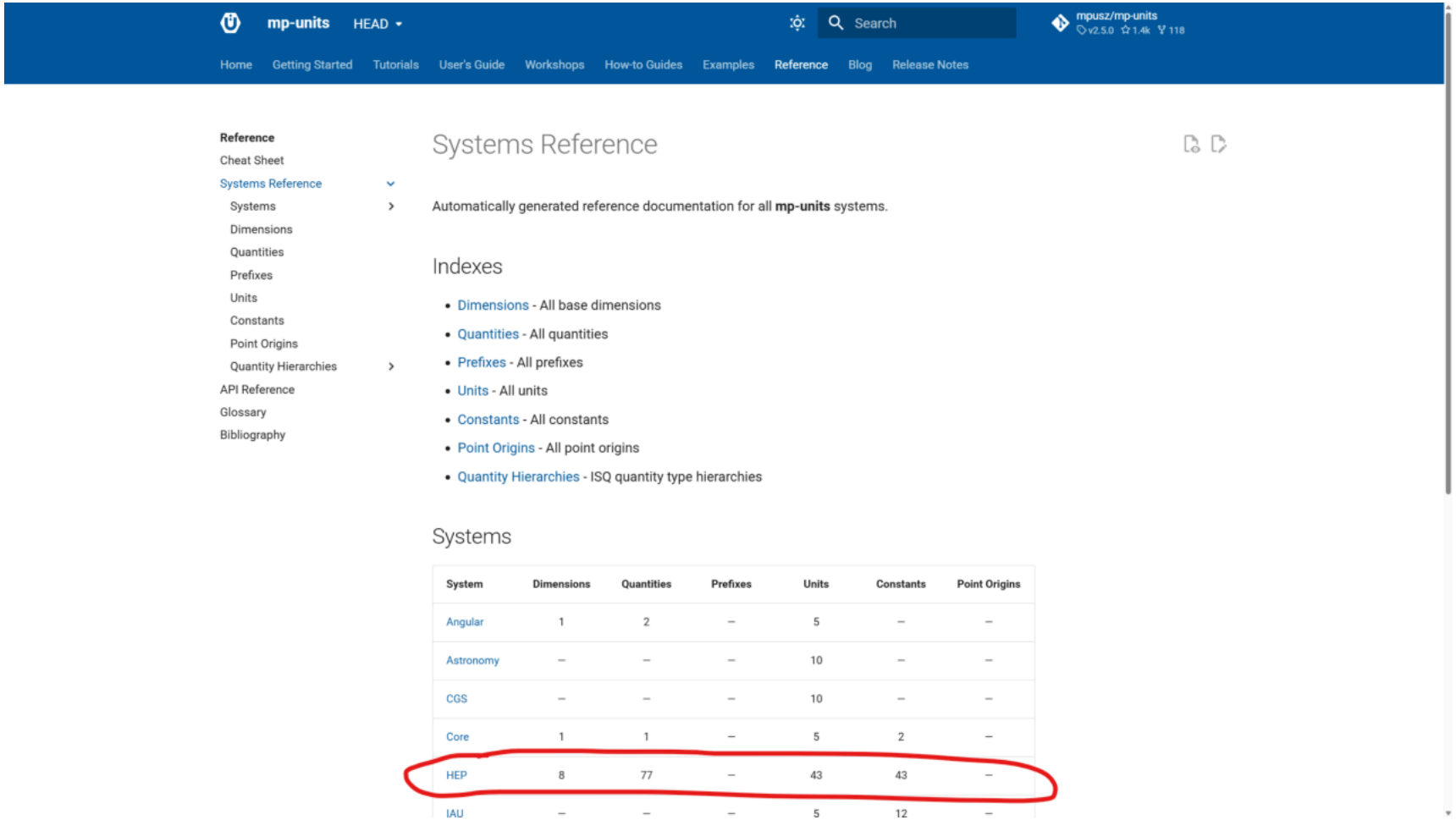
ISQ (INTERNATIONAL SYSTEM OF QUANTITIES)

- *7 base dimensions:* length, mass, time, current, temperature, amount, luminosity
- *Derived:* energy = mass × length^2 / time^2 ($M \cdot L^2 \cdot T^{-2}$)

HEP SYSTEM (HIGH ENERGY PHYSICS)

- **Energy (E)** is a base quantity, not mass!
- **Electric charge (Q)** is base, not current!
- Natural for particle physics: particles characterized by energy, not mass

The HEP System Reference



HEP dimensions and base quantities

```
namespace mp_units::hep {
using namespace ::mp_units::angular;

// dimensions
inline constexpr struct dim_length final : base_dimension<"L"> {} dim_length;
inline constexpr struct dim_time final : base_dimension<"T"> {} dim_time;
inline constexpr struct dim_electric_charge final : base_dimension<"Q"> {} dim_electric_charge;
inline constexpr struct dim_energy final : base_dimension<"E"> {} dim_energy;
inline constexpr struct dim_temperature final : base_dimension<symbol_text{u8"Θ", "O"}> {} dim_temperature;
inline constexpr struct dim_amount_of_substance final : base_dimension<"N"> {} dim_amount_of_substance;
inline constexpr struct dim_luminous_intensity final : base_dimension<"I"> {} dim_luminous_intensity;

// base quantities
inline constexpr struct length final : quantity_spec<dim_length> {} length;
inline constexpr struct duration final : quantity_spec<dim_time> {} duration;
inline constexpr struct electric_charge final : quantity_spec<dim_electric_charge> {} electric_charge;
inline constexpr struct energy final : quantity_spec<dim_energy> {} energy;
inline constexpr struct temperature final : quantity_spec<dim_temperature> {} temperature;
inline constexpr struct amount_of_substance final : quantity_spec<dim_amount_of_substance> {} amount_of_substance;
inline constexpr struct luminous_intensity final : quantity_spec<dim_luminous_intensity> {} luminous_intensity;
}
```

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namespace mp_units::hep {
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inline constexpr struct dim_length final : base_dimension<"L"> {} dim_length;
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inline constexpr struct length final : quantity_spec<dim_length> {} length;
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inline constexpr struct electric_charge final : quantity_spec<dim_electric_charge> {} electric_charge;
inline constexpr struct energy final : quantity_spec<dim_energy> {} energy;
inline constexpr struct temperature final : quantity_spec<dim_temperature> {} temperature;
inline constexpr struct amount_of_substance final : quantity_spec<dim_amount_of_substance> {} amount_of_substance;
inline constexpr struct luminous_intensity final : quantity_spec<dim_luminous_intensity> {} luminous_intensity;
}
```

Should **`angle`** be a strong dimension or dimensionless?

HEP units

```
inline constexpr struct meter final : named_unit<"m", kind_of<length>> {} meter;
inline constexpr struct second final : named_unit<"s", kind_of<duration>> {} second;
inline constexpr struct hertz final : named_unit<"Hz", one / second, kind_of<frequency>> {} hertz;
inline constexpr struct eplus final : named_unit<symbol_text{u8"e⁺", "e+"}, kind_of<electric_charge>> {} eplus;
inline constexpr struct coulomb final : named_unit<"C", mag<6'241'509'074> * mag_power<10, 9> * eplus> {} coulomb;
inline constexpr struct electronvolt final : named_unit<"eV", kind_of<energy>> {} electronvolt;
inline constexpr struct joule final : named_unit<"J", electronvolt * coulomb / eplus> {} joule;
inline constexpr struct gram final : named_unit<"g", mag_ratio<1, 1000> * joule * square(second) / square(meter)> {} gram;

namespace unit_symbols {

inline constexpr auto m = meter;
inline constexpr auto mm = si::milli<meter>;
inline constexpr auto mm2 = square(mm);
inline constexpr auto mm3 = cubic(mm);

inline constexpr auto ns = si::nano<second>;
inline constexpr auto s = second;
inline constexpr auto ms = si::milli<second>;
inline constexpr auto us = si::micro<second>;

inline constexpr auto eV = electronvolt;
inline constexpr auto MeV = si::mega<electronvolt>;
inline constexpr auto GeV = si::giga<electronvolt>;

}
```

HEP System: Derived Quantities

From energy base to physics quantities

```
using namespace mp_units::hep::unit_symbols;

// Mass derived from energy: M = E/c²
QuantityOf<hep::mass> auto m_e = 0.511 * MeV / c2;  // Electron mass

// Momentum: p = E/c
QuantityOf<hep::momentum> auto p = 45.0 * GeV / c;  // Track momentum
```

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// Momentum: p = E/c
QuantityOf<hep::momentum> auto p = 45.0 * GeV / c;  // Track momentum
```

WHY THIS MATTERS FOR CERN?

- Matches *how physicists think* about particles
- *No confusion* about factors of **c**
- Natural for *relativistic calculations*
- Still allows *conversion to SI mass units when needed*

HEP System: Quantity Subkinds

*Why **`proper_time`** and **`coordinate_time`** must be different kinds?*

```
inline constexpr struct proper_time final : quantity_spec<duration, is_kind> {} proper_time;
inline constexpr struct coordinate_time final : quantity_spec<duration, is_kind> {} coordinate_time;

quantity<hep::lorentz_factor[one]> calculate_lorentz_factor(quantity<hep::coordinate_time[ns]> t_lab,
                                                            quantity<hep::proper_time[ns]> tau)
{ return hep::lorentz_factor(t_lab / tau); }
```

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inline constexpr struct proper_time final : quantity_spec<duration, is_kind> {} proper_time;
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quantity<hep::lorentz_factor[one]> calculate_lorentz_factor(quantity<hep::coordinate_time[ns]> t_lab,
                                                            quantity<hep::proper_time[ns]> tau)
{ return hep::lorentz_factor(t_lab / tau); }
```

```
quantity tau   = hep::proper_time(0.385 * ns);    // B⁰ meson rest-frame lifetime
quantity t_lab = hep::coordinate_time(4.2 * ns);  // observed lab time of flight

// ✅ Division across subkinds — produces dimensionless Lorentz factor
quantity gamma = calculate_lorentz_factor(t_lab, tau);  // γ = t_lab / τ ≈ 10.9
// quantity bad = calculate_lorentz_factor(tau, t_lab); // ❌ Compile error!

// ✅ Explicit cast to plain duration when needed (e.g. for legacy APIs)
quantity t = hep::duration(t_lab);    // explicit — intentional
```

- τ: Lorentz-invariant rest-frame lifetime — **frame-independent**
- t_lab: **frame-dependent lab time** — changes with boost

HEP System: Quantity Subkinds

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```
inline constexpr struct proper_time final : quantity_spec<duration, is_kind> {} proper_time;
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quantity<hep::lorentz_factor[one]> calculate_lorentz_factor(quantity<hep::coordinate_time[ns]> t_lab,
                                                            quantity<hep::proper_time[ns]> tau)
{ return hep::lorentz_factor(t_lab / tau); }
```

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quantity tau   = hep::proper_time(0.385 * ns);    // B⁰ meson rest-frame lifetime
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// ✅ Explicit cast to plain duration when needed (e.g. for legacy APIs)
quantity t = hep::duration(t_lab);    // explicit — intentional
```

Subkinds encode physics intent directly into the type system.

HEP System: CODATA Version Control

Ensuring consistent physics constants across frameworks

```
using namespace mp_units::hep::unit_symbols;

// CODATA 2018 (default — matches current Geant4, CLHEP)
std::println("{}", (1. * hep::electron_mass).in(MeV/c2));                    // "0.51099895 MeV/c²"
std::println("{}", (1. * hep::fine_structure_constant).in(one));             // "0.0072973525693"

// CODATA 2022 (matches latest ROOT)
std::println("{}", (1. * hep::codata2022::electron_mass).in(MeV/c2));        // "0.51099895069 MeV/c²"
std::println("{}", (1. * hep::codata2022::fine_structure_constant).in(one)); // "0.0072973525643"

// CODATA 2014 (legacy CLHEP, Gaudi)
std::println("{}", (1. * hep::codata2014::electron_mass).in(MeV/c2));        // "0.5109989461 MeV/c²"
```

HEP System: CODATA Version Control

Ensuring consistent physics constants across frameworks

```
using namespace mp_units::hep::unit_symbols;

// CODATA 2018 (default — matches current Geant4, CLHEP)
std::println("{}", (1. * hep::electron_mass).in(MeV/c2));                    // "0.51099895 MeV/c²"
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// CODATA 2014 (legacy CLHEP, Gaudi)
std::println("{}", (1. * hep::codata2014::electron_mass).in(MeV/c2));        // "0.5109989461 MeV/c²"
```

Compile-time selection ensures:

- No accidental mixing of constant versions
- Clear documentation of which CODATA is used
- Easy migration path as frameworks update

Potential Migration Strategy for HEP Software

John Chapman (ATLAS Offline Software Coordinator) requirements

- Cannot rewrite 2.5M lines of code overnight
- Must prove zero performance impact
- Inside-out: start with unit headers, expand outward
- Evolutionary, not revolutionary

6-Phase Approach

1. Integrate **mp-units** with your code
2. Drop-in **replacements** for unit constant headers
3. **New code** uses typed quantities from the start
4. Wrap existing interfaces with **type-safe overloads**
5. **Refactor internals**; add typed getters; deprecate **`double`** API
6. **Complete** the migration; remove **`_qty`** suffix

Phase 1: Integrate mp-units with your code

Import mp-units framework and HEP system of quantities and units to your project

```
#include <mp-units/math.h>
#include <mp-units/systems/hep.h>

namespace CLHEP {
  using namespace mp_units;

  namespace mp {
    using namespace mp_units::hep;
    using namespace mp_units::hep::unit_symbols;
  }
}
```

- Pick *any namespace* that suits you for storing strong quantities, units, and constants types
- It should *not collide* with your existing identifiers

Phase 2: Drop-In Replacements for Unit Constant Headers

*Replace hand-maintained **`constexpr double`** constants — all existing code continues to compile*

```
// Before: hand-maintained constants — values must be kept in sync manually
namespace CLHEP {
  static constexpr double mm  = 1.;           // Base unit
  static constexpr double cm  = 10. * mm;     // = 10.0
  static constexpr double MeV = 1.;           // Base unit
  static constexpr double GeV = 1.e+3 * MeV;  // = 1000.0
}
```

Phase 2: Drop-In Replacements for Unit Constant Headers

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```
// Before: hand-maintained constants — values must be kept in sync manually
namespace CLHEP {
  static constexpr double mm  = 1.;           // Base unit
  static constexpr double cm  = 10. * mm;     // = 10.0
  static constexpr double MeV = 1.;           // Base unit
  static constexpr double GeV = 1.e+3 * MeV;  // = 1000.0
}
```

```
// After: mp-units derives the values — no manual arithmetic, always correct
namespace CLHEP {
  constexpr double mm  = (1. * mp::mm).numerical_value_in(mp::mm);    // = 1.0  (base unit)
  constexpr double cm  = (1. * mp::cm).numerical_value_in(mp::mm);    // = 10.0
  constexpr double MeV = (1. * mp::MeV).numerical_value_in(mp::MeV);  // = 1.0  (base unit)
  constexpr double GeV = (1. * mp::GeV).numerical_value_in(mp::MeV);  // = 1000.0
}
```

Impact: Existing **`double energy = 50.0 * GeV;`** compiles unchanged

Phase 3: New Code Uses Typed Quantities from the Start

*All newly written functions and classes use **`quantity<...>`** types — legacy code untouched*

```
// New typed function — fully safe, no legacy code touched
quantity<mp::GeV / mp::c> calculate_momentum(quantity<mp::GeV> energy,
                                             quantity<mp::GeV / mp::c2> mass)
{
  return sqrt(pow<2>(energy) - pow<2>(mass * mp::c2)) / mp::c;
}
```

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```
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quantity<mp::GeV / mp::c> calculate_momentum(quantity<mp::GeV> energy,
                                             quantity<mp::GeV / mp::c2> mass)
{
  return sqrt(pow<2>(energy) - pow<2>(mass * mp::c2)) / mp::c;
}
```

```
// Call site — units enforced at compile time
quantity p = calculate_momentum(125. * mp::GeV,                          // ✅ energy
                                4.18 * mp::GeV / mp::c2);                // ✅ mass
// auto p2 = calculate_momentum(125. * mp::GeV, 4.18 * mp::GeV);         // ❌ Compile error: mass expected
// auto p3 = calculate_momentum(4.18 * mp::GeV / mp::c2, 125. * mp::GeV); // ❌ Compile error: arg order wrong
```

Impact: Dimensional safety from day one, no legacy disruption

Phase 4: Wrap Existing Interfaces with Type-Safe Overloads

*Add typed setter overloads — class internals (raw **`double`** members) left completely unchanged*

```
class Particle {
  double m_mass{};    // Still raw double — internals untouched
  double m_width{};
  double m_charge{};
public:
  // Existing double-based interface — unchanged, still works:
  double getMass() const                { return m_mass; }
  double getWidth() const               { return m_width; }
  double getCharge() const              { return m_charge; }

  void setMass(double mass_MeV_per_c2)  { m_mass   = mass_MeV_per_c2; }
  void setWidth(double width_MeV)       { m_width  = width_MeV; }
  void setCharge(double charge_eplus)   { m_charge = charge_eplus; }

  // New typed overloads — convert at the boundary, delegate to existing setters:
  void setMass(quantity<mp::rest_mass[mp::MeV / mp::c2]> mass)    { setMass(mass.numerical_value_in(mp::MeV / mp::c2)); }
  void setWidth(quantity<mp::decay_width[mp::MeV]> width)         { setWidth(width.numerical_value_in(mp::MeV)); }
  void setCharge(quantity<mp::electric_charge[mp::eplus]> charge) { setCharge(charge.numerical_value_in(mp::eplus)); }
};
```

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  void setCharge(quantity<mp::electric_charge[mp::eplus]> charge) { setCharge(charge.numerical_value_in(mp::eplus)); }
};
```

Note: Typed getter overloads cannot be added yet.

Phase 5: Refactor Internals; Add Typed Getters; Deprecate `double` API

*Store typed quantities as members; add **`_qty`** getters; deprecate raw **`double`** API*

```
class Particle {
  quantity<mp::rest_mass[mp::MeV / mp::c2]>  m_mass{};    // Now typed quantities
  quantity<mp::decay_width[mp::MeV]>         m_width{};
  quantity<mp::electric_charge[mp::eplus]>   m_charge{};
public:
  // New typed interface — getters with _qty suffix and setters:
  quantity<mp::rest_mass[mp::MeV / mp::c2]> getMass_qty() const   { return m_mass; }
  quantity<mp::decay_width[mp::MeV]>        getWidth_qty() const  { return m_width; }
  quantity<mp::electric_charge[mp::eplus]>  getCharge_qty() const { return m_charge; }

  void setMass(quantity<mp::rest_mass[mp::MeV / mp::c2]> mass)    { m_mass   = mass; }
  void setWidth(quantity<mp::decay_width[mp::MeV]> width)         { m_width  = width; }
  void setCharge(quantity<mp::electric_charge[mp::eplus]> charge) { m_charge = charge; }

  // Old double interface — now deprecated:
  [[deprecated("Use getMass_qty() instead")]]   double getMass() const   { return m_mass.numerical_value_in(mp::MeV / mp::c2); }
  [[deprecated("Use getWidth_qty() instead")]]  double getWidth() const  { return m_width.numerical_value_in(mp::MeV); }
  [[deprecated("Use getCharge_qty() instead")]] double getCharge() const { return m_charge.numerical_value_in(mp::eplus); }

  [[deprecated("Pass quantity instead")]] void setMass(double mass)      { setMass(mass * mp::MeV / mp::c2); }
  [[deprecated("Pass quantity instead")]] void setWidth(double width)    { setWidth(width * mp::MeV); }
  [[deprecated("Pass quantity instead")]] void setCharge(double charge)  { setCharge(charge * mp::eplus); }
};
```

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  quantity<mp::decay_width[mp::MeV]>         m_width{};
  quantity<mp::electric_charge[mp::eplus]>   m_charge{};
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  // New typed interface — getters with _qty suffix and setters:
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  void setMass(quantity<mp::rest_mass[mp::MeV / mp::c2]> mass)    { m_mass   = mass; }
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  [[deprecated("Pass quantity instead")]] void setMass(double mass)      { setMass(mass * mp::MeV / mp::c2); }
  [[deprecated("Pass quantity instead")]] void setWidth(double width)    { setWidth(width * mp::MeV); }
```

Old call sites trigger deprecation warnings; new code uses the typed API.

Phase 6: Complete the Migration; Remove `_qty` Suffix

Once all call sites use the typed API, remove deprecated overloads and rename getters

```
class Particle {
  quantity<mp::rest_mass[mp::MeV / mp::c2]> m_mass{};
  quantity<mp::decay_width[mp::MeV]>        m_width{};
  quantity<mp::electric_charge[mp::eplus]>  m_charge{};
public:
  // Fully typed interface — deprecated methods removed, _qty suffix removed:
  quantity<mp::rest_mass[mp::MeV / mp::c2]> getMass() const        { return m_mass; }
  quantity<mp::decay_width[mp::MeV]>        getWidth() const       { return m_width; }
  quantity<mp::electric_charge[mp::eplus]>  getCharge() const      { return m_charge; }

  void setMass(quantity<mp::rest_mass[mp::MeV / mp::c2]> mass)     { m_mass   = mass; }
  void setWidth(quantity<mp::decay_width[mp::MeV]> width)          { m_width  = width; }
  void setCharge(quantity<mp::electric_charge[mp::eplus]> charge)  { m_charge = charge; }
};
```

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  quantity<mp::electric_charge[mp::eplus]>  getCharge() const      { return m_charge; }

  void setMass(quantity<mp::rest_mass[mp::MeV / mp::c2]> mass)     { m_mass   = mass; }
  void setWidth(quantity<mp::decay_width[mp::MeV]> width)          { m_width  = width; }
  void setCharge(quantity<mp::electric_charge[mp::eplus]> charge)  { m_charge = charge; }
};
```

This phase is a simple rename: **`sed -i 's/_qty()/()/g'`**.

Performance: Zero Overhead Promise

NO UNITS LIBRARY

```
double ttg_s(double d_m, double v_mps)
{
  return d_m / v_mps;
}
```

MP-UNITS

```
quantity<s> ttg(quantity<m> d, quantity<m/s> v)
{
  return d / v;
}
```

Performance: Zero Overhead Promise

NO UNITS LIBRARY

```
double ttg_s(double d_m, double v_mps)
{
  return d_m / v_mps;
}
```

```
ttg_s(double, double):
  divsd xmm0, xmm1
  ret
```

MP-UNITS

```
quantity<s> ttg(quantity<m> d, quantity<m/s> v)
{
  return d / v;
}
```

```
ttg(quantity<m>, quantity<m/s>):
  divsd xmm0, xmm1
  ret
```

Performance: Zero Overhead Promise

NO UNITS LIBRARY

```
double ttg_s(double d_m, double v_mps)
{
  return d_m / v_mps;
}
```

```
ttg_s(double, double):
  divsd xmm0, xmm1
  ret
```

MP-UNITS

```
quantity<s> ttg(quantity<m> d, quantity<m/s> v)
{
  return d / v;
}
```

```
ttg(quantity<m>, quantity<m/s>):
  divsd xmm0, xmm1
  ret
```

Identical assembly! Units are a compile-time-only concept.

Performance: HEP Unit Conversions

NO UNITS LIBRARY

```
auto foo(double val)
{
  double length = val * CLHEP::mm;
  return length / CLHEP::cm;
}
```

MP-UNITS

```
auto foo(double val)
{
  quantity length = val * mm;
  return length.numerical_value_in(cm);
}
```

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```
auto foo(double val)
{
  double length = val * CLHEP::mm;
  return length / CLHEP::cm;
}
```

```
.LCPI0_0:
  .quad   0x4024000000000000
foo(double):
  divsd   xmm0, qword ptr [rip + .LCPI0_0]
  ret
```

MP-UNITS

```
auto foo(double val)
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  quantity length = val * mm;
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  return length.numerical_value_in(cm);
}
```

Performance: HEP Unit Conversions

NO UNITS LIBRARY

```
auto foo(double val)
{
  double length = val * CLHEP::cm;
  return length / CLHEP::cm;
}
```

```
.LCPI0_0:
  .quad   0x4024000000000000
foo(double):
  movsd   xmm1, qword ptr [rip + .LCPI0_0]
  mulsd   xmm0, xmm1
  divsd   xmm0, xmm1
  ret
```

MP-UNITS

```
auto foo(double val)
{
  quantity length = val * cm;
  return length.numerical_value_in(cm);
}
```

```
foo(double):
  ret
```

Standardization: Path to C++ Standard

ISO C++ PAPERS

- P1935: A C++ Approach to Physical Units
- P2980: A motivation, scope, and plan for a physical quantities and units library
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CURRENT STATUS

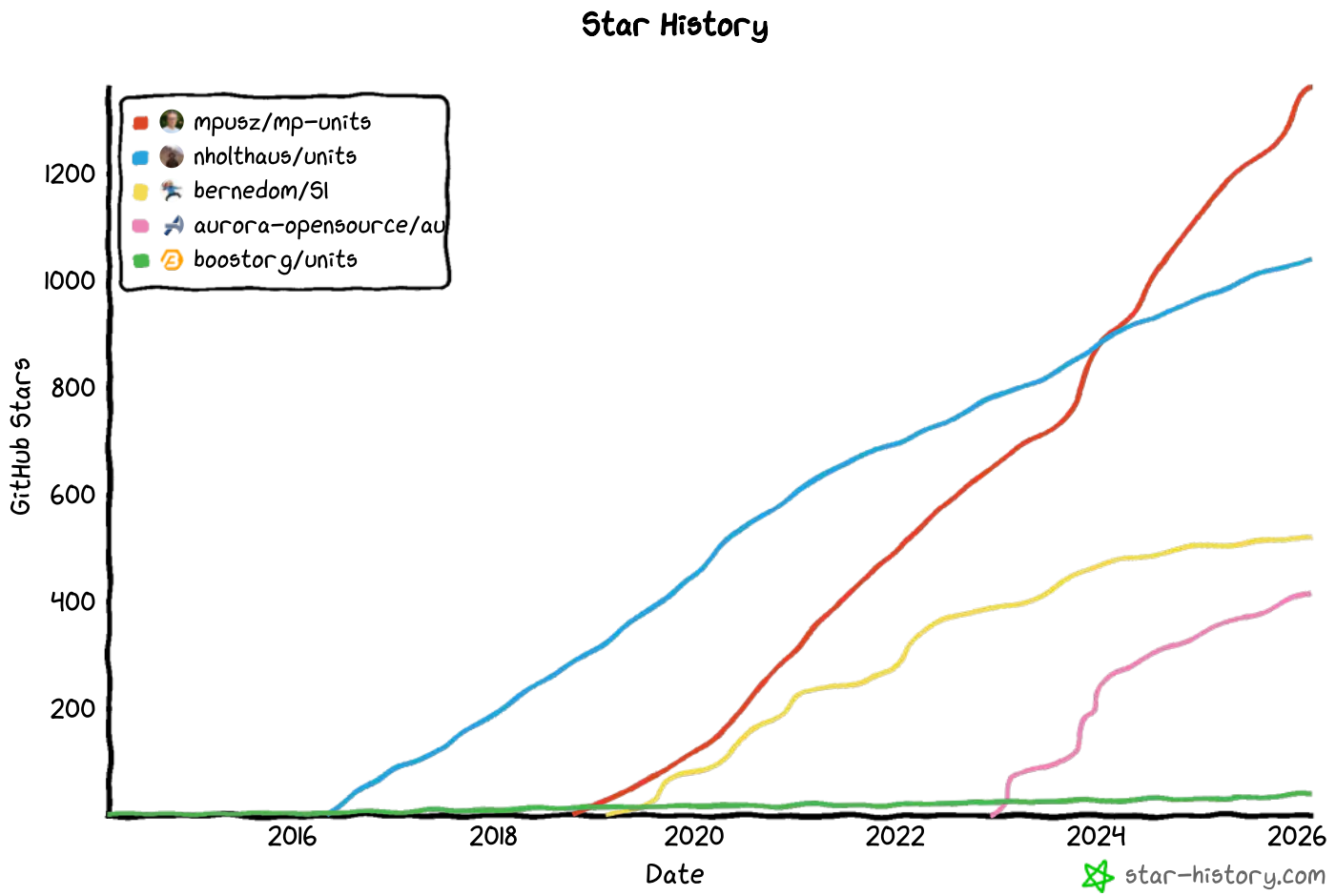
- Presented to LEWG (Library Evolution Working Group)
- Positive feedback from the Committee
- **mp-units** used as reference implementation for discussion
- Target: C++29

Standardization: Path to C++ Standard

WHY THIS MATTERS FOR HEP?

- Investing in future standard library
- Broad C++ community review and feedback
- Eventually part of every compiler
- Skills transferable beyond HEP

Comparison: mp-units vs. Alternatives



Comparison: mp-units vs. Alternatives

SAFETY FEATURE	MP-UNITS (CURRENT)	MP-UNITS (V3 PLANNED)	BOOST.UNITS	NHOLTHAUS	SI	AU
Minimum Requirement	C++20	C++20	C++11/14	C++14	C++17	C++14
Dimension Safety	✅ Full	✅ Full	✅ Full	✅ Full	✅ Full	✅ Full
Unit Safety	✅ Full	✅ Full	⚠️ Partial	⚠️ Partial	⚠️ Limited	✅ Full
Representation Safety	✅ Strong	✅ Strong	⚠️ Partial	⚠️ Partial	⚠️ Limited	⭐ Strong+
Quantity Kind Safety	✅ Full	✅ Full	⚠️ Partial	❌ None	❌ None	❌ None
Quantity Safety	⚠️ Partial	✅ Full	❌ None	❌ None	❌ None	❌ None
Affine Space Safety	✅ Full	✅ Full	⚠️ Partial	❌ None	❌ None	✅ Full
HEP System	✅ Built-in	✅ Built-in	❌ No	❌ No	❌ No	❌ No

Discussion & Next Steps

IMMEDIATE ACTIONS

- **Prototype Phases 1–2**: Namespace setup + drop-in header replacement in Gaudi/CLHEP
- **Benchmark**: Verify zero overhead in ATLAS software stack
- **Pilot Project**: Pick one subsystem (geometry or DAQ) for Phase 3 trial

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MEDIUM-TERM GOALS

- Develop ROOT/Geant4 bridge with type safety
- Train developers (tutorials, documentation)
- Establish coding guidelines for typed interfaces

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MEDIUM-TERM GOALS

- Develop ROOT/Geant4 bridge with type safety
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- Establish coding guidelines for typed interfaces

LONG-TERM VISION

- Full typed API across ATLAS/CMS software
- Contribution back to mp-units for HEP community
- Part of C++ standard (C++29)

Key Takeaways

1. **Current CERN code** collapses 7+ physical dimensions into 0-dimensional **`double`**
2. **mp-units** restores those dimensions at **compile-time** with **zero runtime cost**
3. **HEP system** uses energy as base quantity, matching how physicists think
4. **6 levels of safety** catch bugs that training alone cannot prevent
5. **Migration strategy** proven: inside-out, evolutionary not revolutionary
6. **Real-world bugs** (G4Trap, ROOT/Geant4 bridges, CODATA drift) are **preventable**

Key Takeaways

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The compiler can protect scientific integrity—if we give it the information!


CAUTION
Programming
is addictive
(and too much fun)